AWG Wire Gauge Chart

Diameter, cross-sectional area and circular mils for 4/0 to 40 AWG

AWG 4/0–40
44 gauges
1 page

AWG 4/0 – 40	DIAMETER INCH	DIAMETER MM	AREA MM²	CIRCULAR MILS CMIL	NEAREST IEC MM²
4/0	0.4600	11.684	107.2	211,600	95
3/0	0.4096	10.405	85.03	167,800	95
2/0	0.3648	9.266	67.43	133,100	70
1/0	0.3249	8.251	53.48	105,500	50
1	0.2893	7.348	42.41	83,690	35
2	0.2576	6.544	33.63	66,370	35
3	0.2294	5.827	26.67	52,630	25
4	0.2043	5.189	21.15	41,740	25
5	0.1819	4.621	16.77	33,100	16
6	0.1620	4.115	13.30	26,250	16
7	0.1443	3.665	10.55	20,820	10
8	0.1285	3.264	8.366	16,510	10
9	0.1144	2.906	6.634	13,090	6
10	0.1019	2.588	5.261	10,380	6
11	0.0907	2.305	4.172	8,234	4
12	0.0808	2.053	3.309	6,530	4
13	0.0720	1.828	2.624	5,178	2.5
14	0.0641	1.628	2.081	4,107	2.5
15	0.0571	1.450	1.650	3,257	1.5
16	0.0508	1.291	1.309	2,583	1.5
17	0.0453	1.150	1.038	2,048	1.0
18	0.0403	1.024	0.8230	1,624	0.75
19	0.0359	0.912	0.6527	1,288	0.75
20	0.0320	0.812	0.5176	1,022	0.5
21	0.0285	0.723	0.4105	810.1	0.5
22	0.0253	0.644	0.3255	642.4	0.5
23	0.0226	0.573	0.2582	509.5	0.5
24	0.0201	0.511	0.2047	404.0	0.5
25	0.0179	0.455	0.1624	320.4	0.5
26	0.0159	0.405	0.1288	254.1	0.5
27	0.0142	0.361	0.1021	201.5	0.5
28	0.0126	0.321	0.08098	159.8	0.5
29	0.0113	0.286	0.06422	126.7	0.5
30	0.0100	0.255	0.05093	100.5	0.5
31	0.0089	0.227	0.04039	79.70	0.5
32	0.0080	0.202	0.03203	63.21	0.5
33	0.0071	0.180	0.02540	50.13	0.5
34	0.0063	0.160	0.02014	39.75	0.5
35	0.0056	0.143	0.01597	31.52	0.5
36	0.0050	0.127	0.01267	25.00	0.5
37	0.0045	0.113	0.01005	19.83	0.5
38	0.0040	0.101	0.007967	15.72	0.5
39	0.0035	0.090	0.006318	12.47	0.5
40	0.0031	0.080	0.005010	9.888	0.5

11 AWG HAS FOUR PUBLISHED ANSWERS

All four are correct. They differ only in where the rounding chain stops:

formula 0.09074200 in × 25.4 → **2.305 mm**
four-place nominal 0.0907 × 25.4 → **2.304 mm**
three-place nominal 0.090 × 25.4, two places → **2.29 mm**

22 AWG is a different case: Engineering ToolBox prints 0.0254 in where the formula gives 0.0253, and two other sources agree with the formula. That one is a typo, not a convention.

WHERE THESE NUMBERS COME FROM

Definition — $d = 0.005 \times 92^{(36-n)/39}$ in. 4/0 to 1/0 are $n = -3 \ldots 0$. ASTM B258 is this rounded to four places.

Area — $\pi/4 \times d^2$. Second-order, so every rounding choice compounds twice.

Circular mils — (d in thousandths of an inch)². Exact by construction.

Series — every 6 gauges the diameter doubles (2.00503×); every 3 gauges the area doubles (2.00503×).

IEC column — nearest standard metric conductor. Cable is ordered in mm², so use this column to buy, never as an exact conversion.

Shaded — 6 to 14 AWG, most branch-circuit and feeder work.